

Foundation Models Used Acknowledgment

Group: MA_AppliedSession01_E

Unit: FIT2099 S1 2026 Assignment: Assignment 2

Foundation Models Used: ChatGPT / Claude / Gemini

Specific Contributions and Guided Usage

We used the foundation models mentioned above in a guided fashion to assist with the following technical components of the assignment:

1. Requirement 1: Economy System and Supercomputer Transactions

Purpose: Reviewing the buying and selling system through the Supercomputer, including Sellable and Purchasable item contracts, wallet credit handling, and access card level design.

AI Contribution: We consulted the model mainly to review the REQ1 design and check whether the structure aligned with object-oriented principles such as DRY, SRP, OCP, DIP, and KISS. The model was used to help verify that the Supercomputer transaction flow, Wallet/CreditHolder logic, and AccessCard/AccessLevel design were clear, maintainable, and consistent with the requirement.

2. Requirement 3: Spatial Query Utility (SpatialSearch)

Purpose: Centralizing the logic for detecting actors in adjacent tiles to fulfill the DRY principle.

AI Contribution: We consulted the model for best practices in isolating grid-traversal logic from game-logic classes. The AI helped guide the implementation of the SpatialSearch utility class, specifically the getNearbyWorkers and getNearbyActors methods, ensuring that Requirement 3 (Flora triggers) and Requirement 4 (Creature side effects) and Requirement 5 (Stateful creatures) used a consistent detection algorithm.

3. Requirement 3: State Pattern for Mutated Flora Lifecycle

Purpose: Implementation of an evolving growth system for Fleshy and Warper trees.

AI Contribution: The model provided guidance on implementing the State Pattern to handle the transitions between Sprout, Sapling, and Mature stages. It assisted in designing the TreeStage interface and ensured that the AbstractTree class could polymorphically delegate its behavior to the current stage, fulfilling the Open-Closed Principle (OCP).

4. Requirement 4: Parasite Targeting and Infection Lifecycle

Purpose: Developing unique host reactions for Workers, Undead, and inorganic items.

AI Contribution: Guided the logic for the "Targeting Priority Algorithm" and the "Living Hive" lifecycle for infected Workers. It assisted in documenting the application of the Interface Segregation Principle (ISP) via the Infectable interface, ensuring distinct behaviors for biological hosts versus fuel-based items like Lanterns and Cookies.

5. **Requirement 4: Strategy Pattern for Spawning Logic**

Purpose: Implementing map-specific spawning behaviors for Holes and Vents across different locations.

AI Contribution: Consulted the model to refine technical justifications for using the Strategy Pattern to handle map-specific spawning without modifying core class logic. The AI also assisted in brainstorming the "1% expansion" mechanic and the inheritance of capabilities, ensuring the logic adhered to the Open-Closed Principle (OCP).

6. **Requirement 5: Strategy Pattern for Teleportation Devices**

Purpose: Implementing three distinct teleportation mechanisms (Teleportation Tube, Alien Cube, Magic Circle) with different destination selection algorithms and side effects, while maintaining a unified execution flow.

AI Contribution: Consulted the model to refine technical justifications for using the Strategy Pattern to encapsulate device-specific teleportation behaviours without modifying the core TeleportAction class. The also model helped brainstorm the 50% malfunction mechanic for the Teleportation Tube, the 3-random-destination generation for the Alien Cube

Acknowledgment of Responsibility

We acknowledge that while foundation models were used to guide the implementation of these specific logic blocks, we are entirely responsible for the final code submitted. We have verified all citations and code logic for accuracy.